Cal Poly Rec Center Usage Prediction

Final Report

Advanced Machine Learning Final Project

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GitHub repository:
github.com/tyleroberts4/Advanced-Machine-Learning (see the `rec_center/` folder for notebooks, figures, and code)

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1. Problem Statement

The Cal Poly Recreation Center is one of the most heavily used student facilities on campus, yet students often have no reliable way to know how crowded it will be before they arrive. Long lines, full weight rooms, and packed cardio areas create frustration and can discourage students from using a resource they already pay for.

This project addresses a practical question: **Can we predict how busy the Rec Center (and specific areas within it) will be at a given date and time?**

We built two complementary prediction tools:

1. **Regression model** - estimates exact utilization (occupancy relative to capacity, on a 0-1 scale).
2. **Classification model** - labels each time slot as **low**, **medium**, or **high** usage for easy student-facing recommendations.

These predictions can support student planning ("When is the best time to go?"), Rec Center staffing decisions, and future integration with a mobile app or website.

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2. Data Description

Source and coverage

The data comes from **Occuspace**, an occupancy-analytics platform that estimates how many people are in tracked spaces. We used Cal Poly Rec Center exports collected every **30 minutes** from **6:00 AM to 11:30 PM**, spanning **May 2023 through April 2026**.

After cleaning, the modeling dataset contains **223,283 observations** across **six locations**:

Location	Role
Rec Center (overall)	Whole-facility crowding
1st Floor	General floor traffic
2nd Floor	General floor traffic
Lower Exercise Room	Strength/cardio area
Upper Exercise Room	Strength/cardio area
Track Exercise Room	Track and adjacent equipment

Key variables

- **Average Utilization** (primary target): average occupancy divided by listed capacity. A value of 0.50 means the space is about half full; 1.0 means at capacity.
- **Time predictors**: hour of day, day of week, month, weekend indicator.

- **Location**: which area of the Rec Center is being measured.
- **Capacity**: maximum listed occupancy for each space (used for context, not as a standalone predictor for utilization).

Data quality notes

- No missing values in the original export columns.
- **216 duplicate** location/timestamp pairs were removed.
- **22,069 rows** had average utilization above 1.0 (Occuspace estimated more people than listed capacity). We retained these observations but **capped utilization at 1.0** for modeling, since values above 100% likely reflect sensor or capacity-listing error rather than a fundamentally different crowding pattern.

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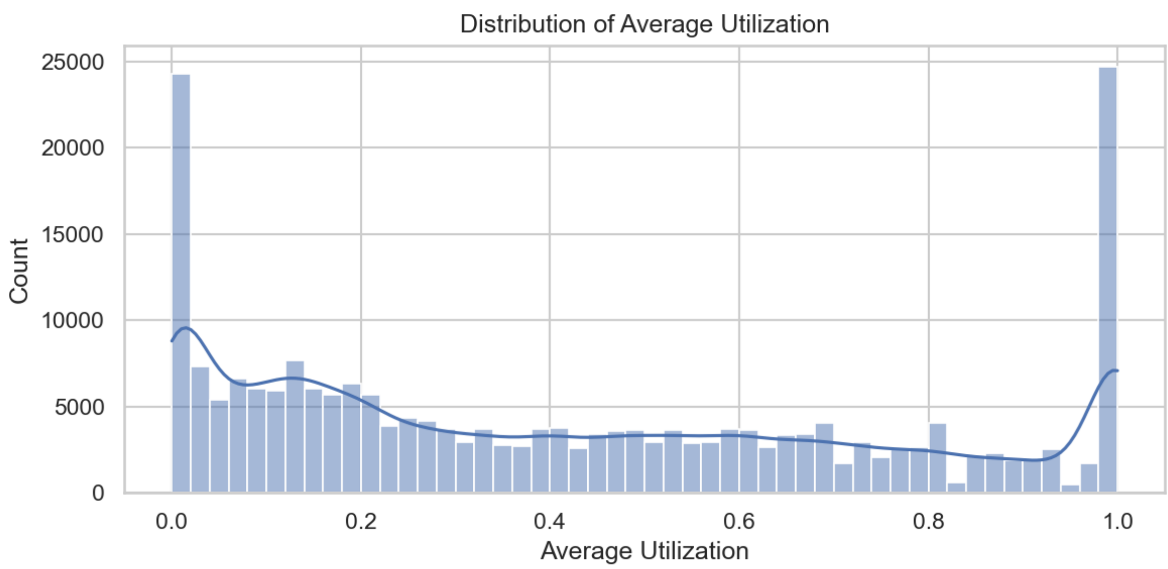
3. Exploratory Analysis

Our analysis of three years of usage data revealed clear, actionable patterns.

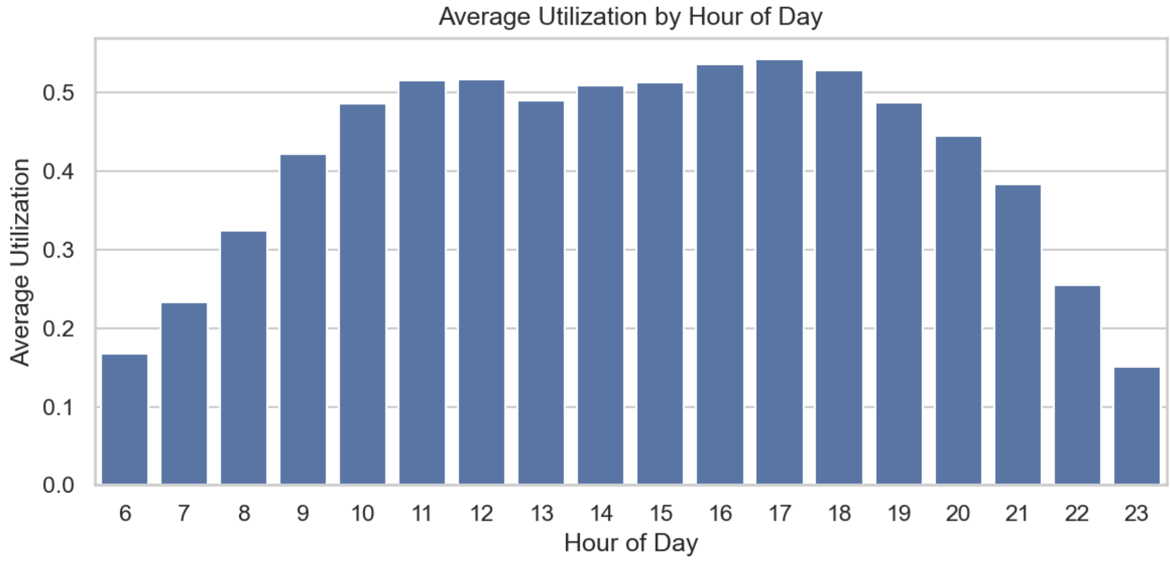
Overall utilization

Average utilization across all locations and times is **0.44** (44% of capacity), with a median of **0.35**. The distribution is right-skewed: most intervals are moderately busy, but a meaningful share of periods are very crowded.



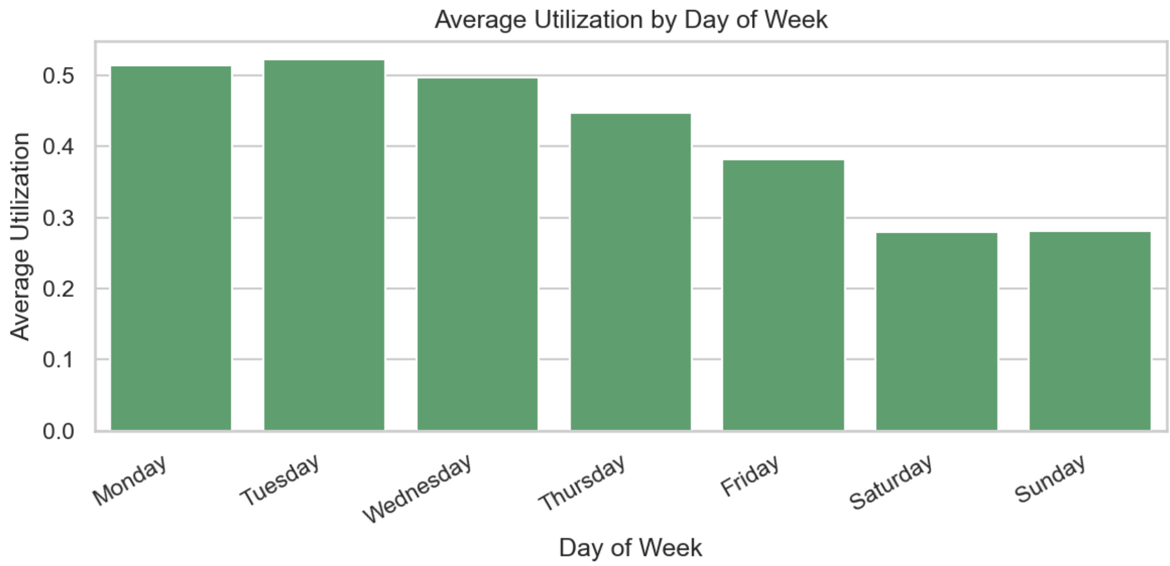
Time-of-day patterns

Usage is lowest in the early morning and late evening. Crowding builds through the afternoon and peaks between **4:00 PM and 6:00 PM**, with **5:00 PM** showing the highest average utilization.



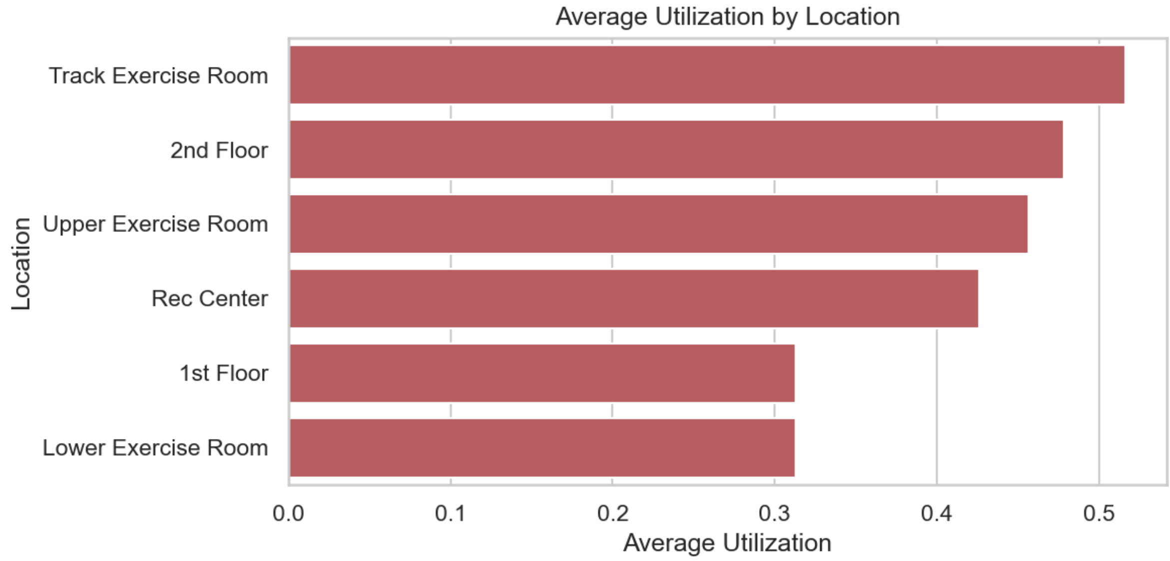
Day-of-week patterns

Monday and Tuesday are the busiest days, followed by Wednesday. **Saturday and Sunday** are consistently quieter - roughly 15-20% lower average utilization than midweek days. Weekday labels in the chart follow the standard Monday-through-Sunday calendar order.



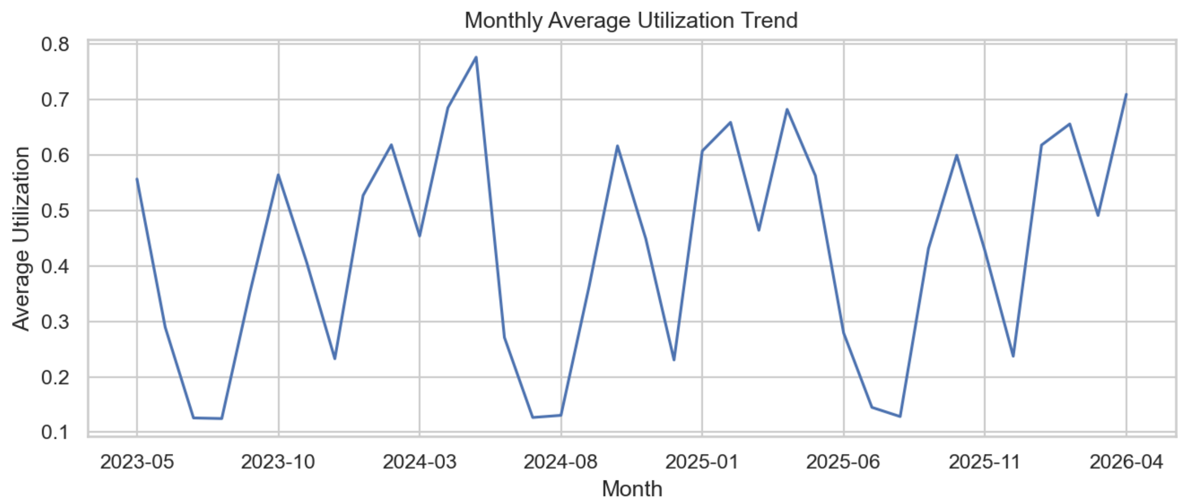
Location differences

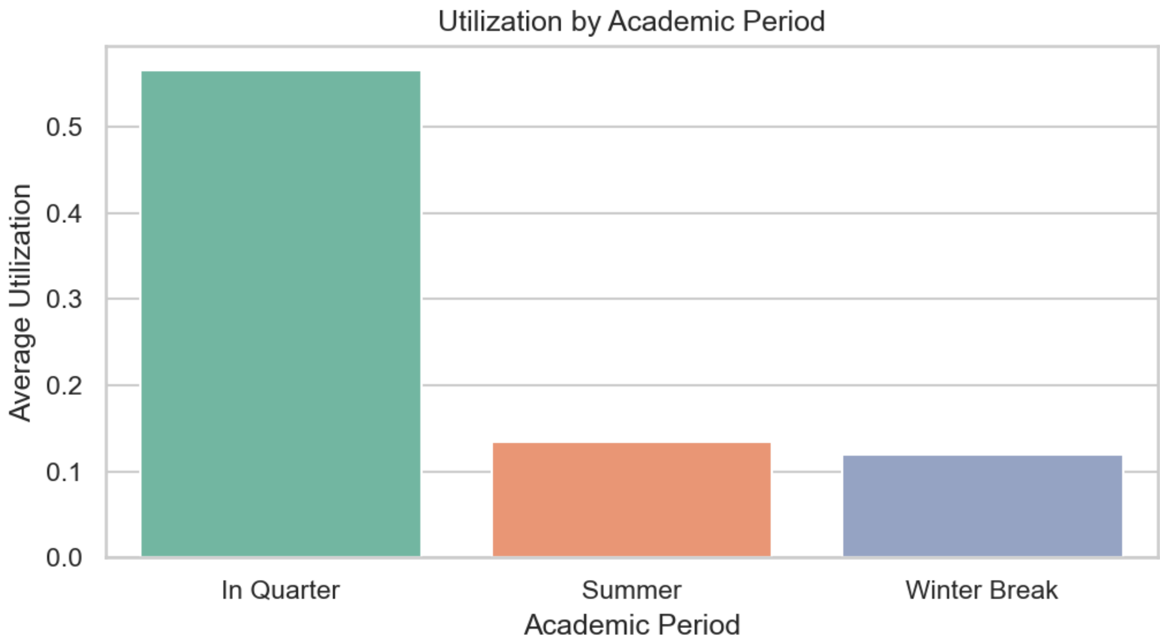
Not all areas are equally crowded. The **Track Exercise Room** averages **0.58** utilization, followed by the **2nd Floor** (0.52) and **Upper Exercise Room** (0.49). The **1st Floor** and **Lower Exercise Room** are the least crowded at roughly **0.31** each - useful alternatives when popular areas are full.



Seasonal and academic calendar effects

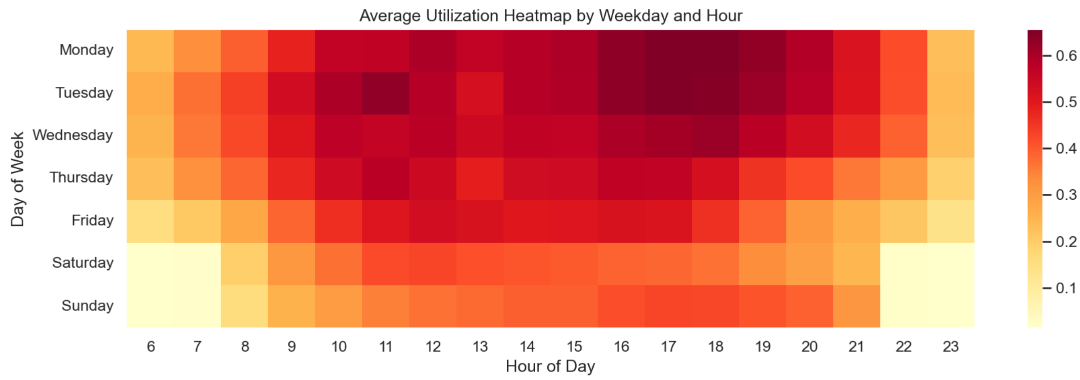
Monthly trends show visible dips during **summer** and **winter break**, and higher usage during active academic quarters. August 2023 and summer months consistently show lower utilization, while May 2024 showed the highest monthly average - likely reflecting pre-finals and end-of-quarter activity.





Weekday-hour interactions

A heatmap of weekday versus hour confirms that **weekday late afternoons** are the hottest windows, while **weekend mornings** and **late evenings** are consistently quieter. These interaction patterns motivated our use of flexible, nonlinear models rather than simple linear formulas.



4. Modeling Approach

How we trained and tested

We treated this as a **forecasting problem**: the model learns from past usage and predicts future periods it has never seen. Data was split by time:

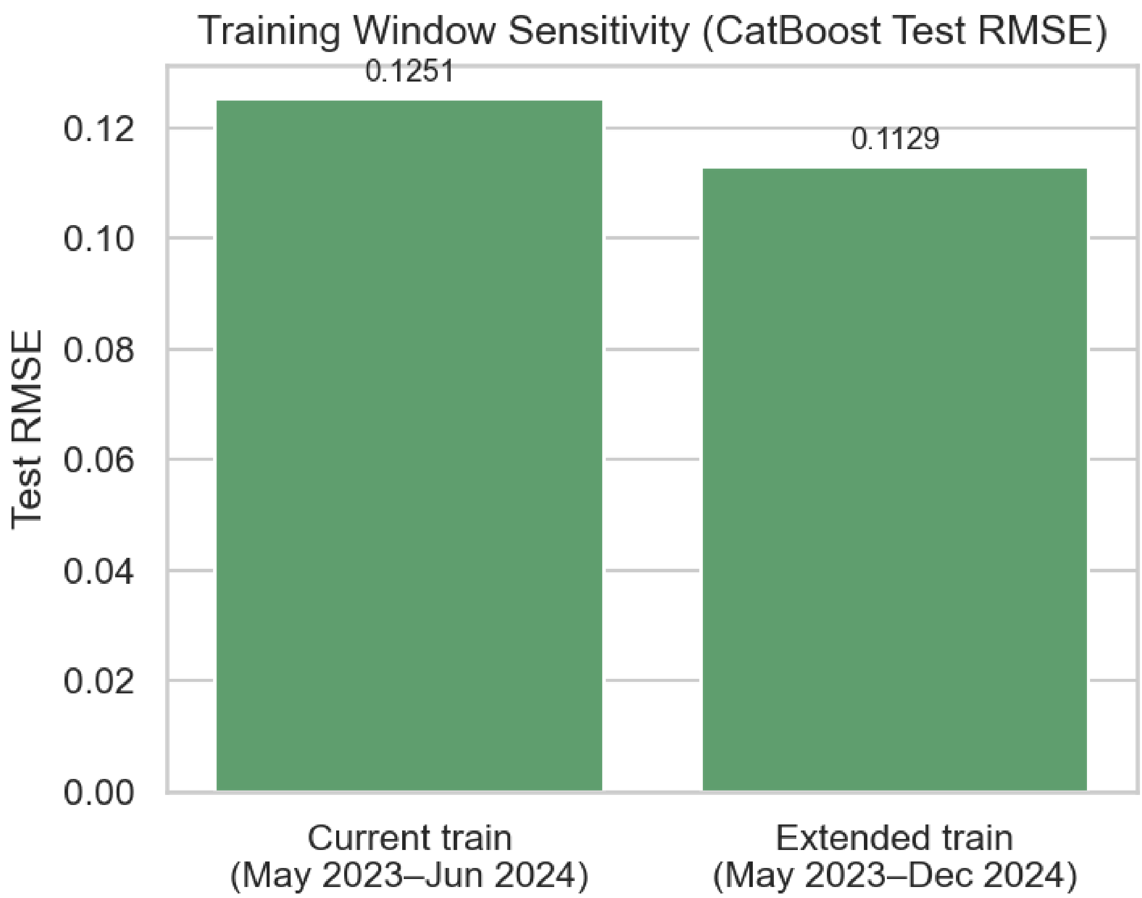
Split	Calendar span	Approx.	Observations	Purpose
Training	May 2023 - June 2024	~14	82,679	Learn historical patterns

Validation	July - Dec 2024	6	39,596	Tune and confirm model settings
Test	Jan 2025 - Apr 2026	~16	101,008	Final unbiased evaluation

We deliberately avoided random train/test splits, which would mix future data into training and overstate accuracy.

Why is the test window longer than training? The test period includes all data after validation through the end of our export (April 2026). The test set is larger than training because all post-validation data through April 2026 was reserved for final evaluation rather than discarded. Holding out the full January 2025 to April 2026 period gives a large, future-facing test set that covers multiple academic terms and seasonal patterns. The six-month validation window sits between train and test so hyperparameters are chosen on recent-but-not-future data.

Would a longer training window help? As a supplementary check, extending training through December 2024 improved test RMSE slightly (**0.125 to 0.113**). We report results from the standard train/validation/test split because it preserves a dedicated tuning window; in production, models would be retrained on all available history after settings are chosen.



Features used

Beyond raw time and location fields, we engineered:

- **Weekend indicator** and **cyclical time encodings** (sin/cos transforms for hour and month)

- **Academic calendar flags**: summer, winter break, spring break, and finals week (derived from Cal Poly's academic calendar)

We excluded peak occupancy and peak utilization from predictors because they describe the same 30-minute window as the target and would inflate performance artificially.

Models compared

Regression (predict exact utilization):

Model	Description
Historical mean baseline	Average utilization by location × weekday × hour
Ridge regression	Linear model with regularization
Random Forest	Ensemble of decision trees
LightGBM / CatBoost	Gradient boosting (state-of-the-art for tabular data)
Keras MLP	Neural network with two hidden layers
Stacking ensemble	Combines LightGBM, CatBoost, and MLP via a meta-learner

Classification (predict low / medium / high):

Category	Threshold
Low	Utilization below 0.30
Medium	0.30 - 0.60
High	Above 0.60

These cutoffs align with the data median (0.35) and mean (0.44), giving intuitive "quiet / moderate / crowded" labels. These thresholds were chosen to create practical student-facing categories: below 30% generally indicates a quiet facility, 30-60% reflects moderate use, and above 60% represents periods where students are more likely to experience crowding.

Models compared: logistic regression, random forest, LightGBM, CatBoost, and a Keras neural network classifier.

Hyperparameter tuning

How we searched for settings. For regression, Ridge, Random Forest, LightGBM, and CatBoost each used **RandomizedSearchCV** with 3-fold cross-validation on the training set (10-12 random draws per model). Ridge tuned regularization strength (alpha from 0.001 to 100 on a log scale). Random Forest tuned tree count, depth, and minimum leaf size. LightGBM tuned number of trees, learning rate, leaf count, and row subsampling. CatBoost tuned tree depth, learning rate, iteration count, and L2 regularization. The Keras neural network used **Optuna** for 12 trials over two hidden-layer sizes, dropout, and learning rate. The stacking ensemble used a fixed Ridge meta-learner (alpha = 1.0) fit on validation predictions from the three base models. Classification models followed the same RandomizedSearchCV approach (8-10 iterations per model) but were scored by **macro-F1** instead of RMSE. All final models were evaluated once on the held-out test period.

Selected hyperparameters (final models):

- Ridge: alpha = 54.6
- Random Forest: 300 trees, minimum leaf size 5, unlimited depth
- LightGBM: 600 trees, learning rate 0.05, 63 leaves, subsample 0.9
- CatBoost (regression): depth 10, learning rate 0.05, 500 iterations, L2 regularization 5

- Keras MLP (regression): 128 -> 16 units, dropout 0.29, learning rate 0.0045
- Stacking ensemble meta weights: LightGBM 0.21, CatBoost 0.78, MLP 0.05
- CatBoost (classification): depth 8, learning rate 0.1, 500 iterations

5. Results and Model Comparison

Regression results (test set)

Model	RMSE	MAE	R^2
Stacking ensemble	0.123	0.089	0.872
CatBoost	0.125	0.088	0.867
LightGBM	0.128	0.088	0.862
Random Forest	0.134	0.090	0.848
Keras MLP (neural network)	0.143	0.100	0.826
Ridge regression	0.219	0.175	0.593
Historical mean baseline	0.289	0.247	0.291

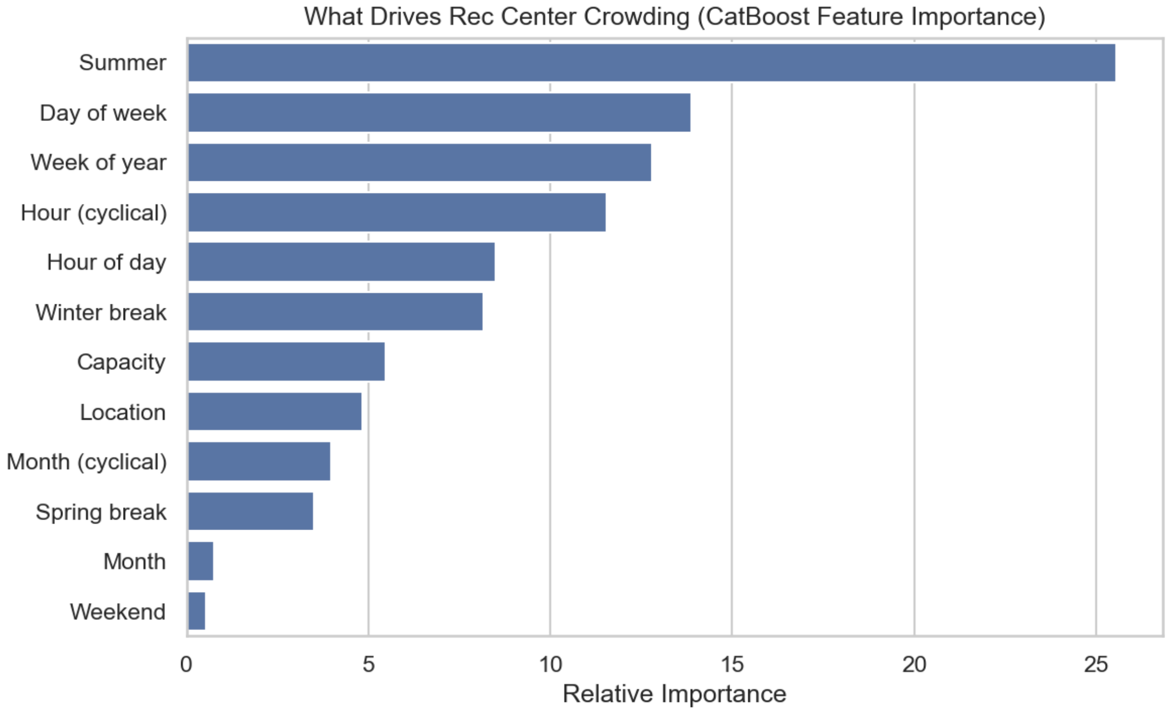
How to read these metrics:

- **RMSE (root mean squared error) -** our primary metric. It penalizes large misses heavily, which matters most during peak hours when students care most about accuracy. An RMSE of 0.123 means typical prediction errors are roughly 12 percentage points of utilization.
- **MAE (mean absolute error)** - easier to interpret: on average, predictions are off by about **9 percentage points** for the best models.
- **R^2** - share of variance explained. Our best model explains about **87%** of utilization variation on unseen future data.

On the validation set (July-December 2024), CatBoost reached RMSE **0.130** and LightGBM **0.132**, closely matching test performance - a sign that tuning generalized rather than overfitting to a single split.

What drives crowding?

Beyond overall accuracy, we asked which inputs the best tree model (CatBoost) relies on most. The chart below aggregates one-hot location columns into a single **Location** bar.



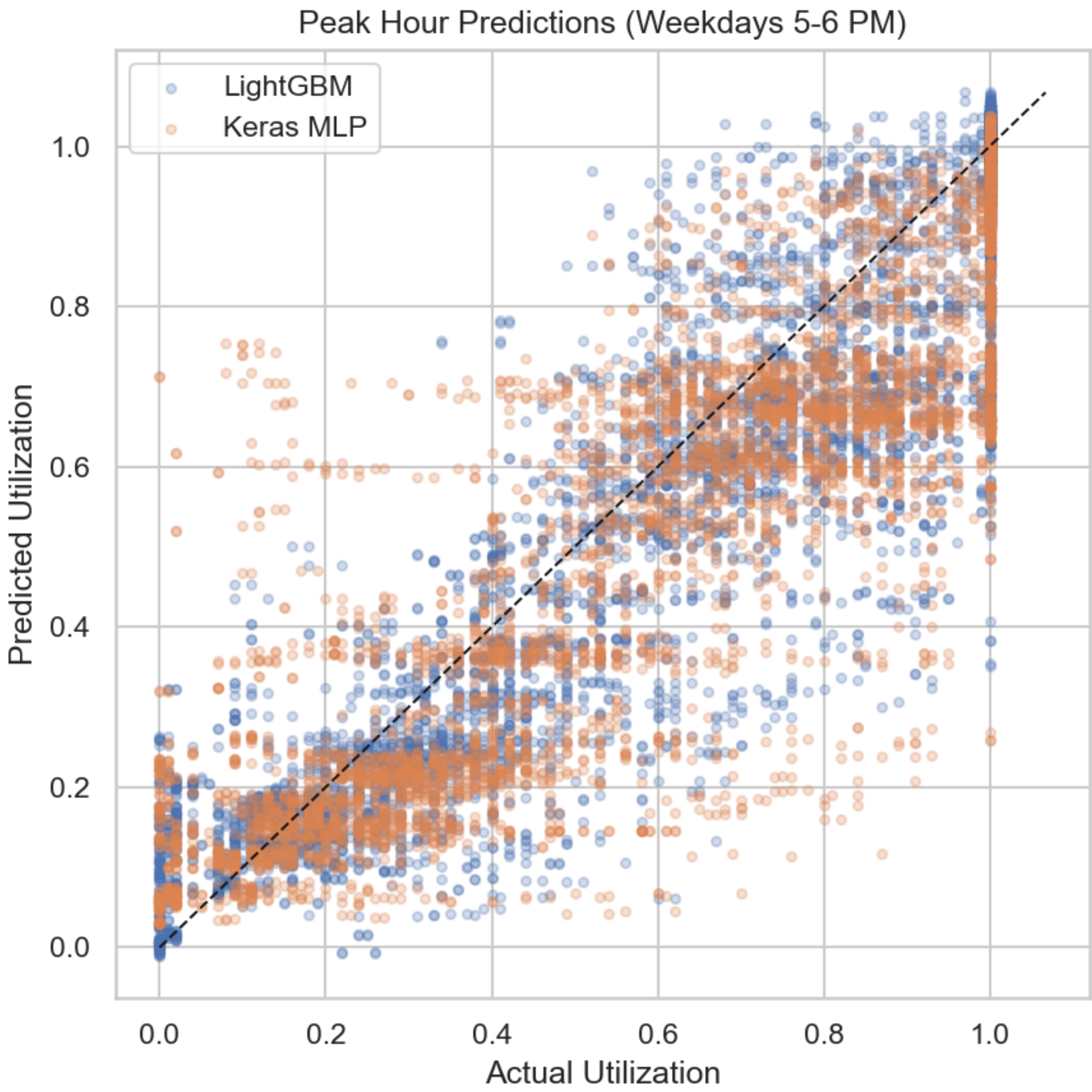
Takeaways:

- **Time and calendar context dominate.** Day of week, week of year, and hour-related features (raw hour plus cyclical encodings) are among the strongest predictors - consistent with EDA showing weekday 4-6 PM peaks.
- **Academic calendar flags matter.** Summer, winter break, and related flags rank highly; a separate ablation (below) confirms that adding calendar features materially improves accuracy.
- **Location still matters.** Which area of the Rec Center is being measured is a meaningful driver, even after accounting for time - aligning with the Track Room and 2nd Floor running much busier than the 1st Floor.
- **Capacity plays a supporting role.** Listed room capacity adds context but is less important than when and where the reading occurs.

Neural network vs. non-neural network

Tree-based models (LightGBM, CatBoost, Random Forest) outperformed the neural network on this dataset. This is a common finding for structured tabular data with strong categorical interactions (location × hour × weekday). The neural network still beat linear baselines by a wide margin but did not justify its added complexity over gradient boosting alone.

The stacking ensemble achieved a modest improvement over the best individual model (RMSE 0.123 vs. 0.125 for CatBoost alone), suggesting some benefit from combining diverse model types, though the gain is small relative to the engineering effort. Although the stacking ensemble achieved the lowest RMSE, **CatBoost is the recommended production model** because it performs nearly as well while being simpler to maintain and explain.

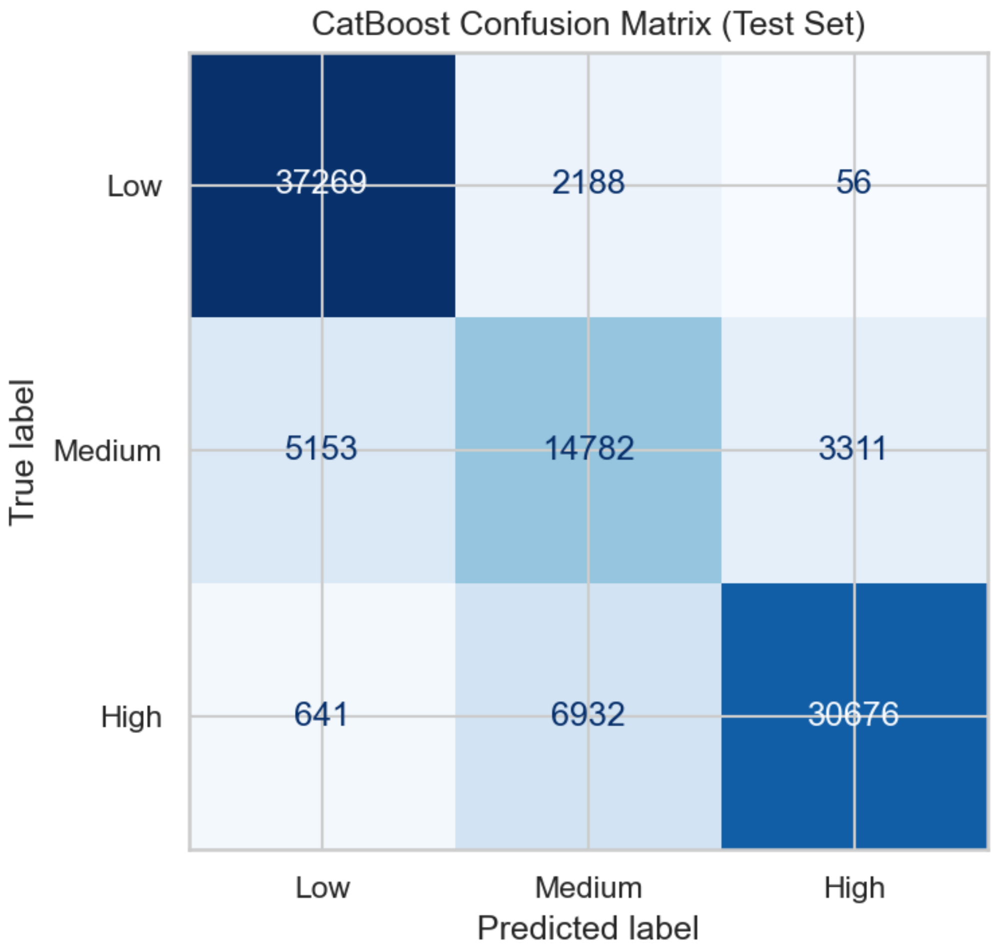


Classification results (test set)

Model	Accuracy	Macro-F1
CatBoost	81.9%	0.793
LightGBM	81.3%	0.787
Random Forest	81.3%	0.784
Keras MLP	80.4%	0.779
Logistic regression	70.9%	0.658

Macro-F1 is our primary classification metric because it weights low, medium, and high usage classes equally - important since high-crowd periods are less frequent but most valuable to predict correctly.

CatBoost correctly classifies usage level about **82%** of the time on held-out future data, with strong performance across all three categories.



Most classification errors occur between neighboring categories, especially medium vs. high, which is less concerning than confusing high-usage periods with low-usage periods.

Feature impact: academic calendar (ablation)

As complementary evidence to the global importance chart, we trained LightGBM with and without academic calendar flags. Adding summer, breaks, and finals indicators improved test RMSE from **0.145 to 0.128 -** confirming that campus schedule effects are real and worth modeling even when hour and location already explain most variation.

6. Conclusions and Recommendations

Key findings

1. **Rec Center usage is highly predictable** from time, location, and calendar context. Our best model achieves ~9-point average error on a 0-1 scale for future time periods.
2. **Peak crowding is concentrated** on weekday afternoons (4-6 PM), especially Monday through Wednesday.
3. **Location matters significantly.** The Track Exercise Room and 2nd Floor run 60-80% higher utilization than the 1st Floor and Lower Exercise Room.

4. **Summer and breaks are reliably quieter**, making them good times for students who prefer less crowded workouts.

5. **CatBoost is the recommended deployment model.** The stacking ensemble marginally outperforms it on RMSE, but CatBoost offers nearly equivalent accuracy with less complexity; neural networks add further cost without meaningful gains on this dataset.

Recommendations for Rec Center staff and students

For students:

- Visit **early morning (before 10 AM)** or **weekends** for the lowest crowding.
- Avoid **Monday-Wednesday, 4:00-6:00 PM** if possible.
- If the Track Room or 2nd Floor is full, try the **Lower Exercise Room or 1st Floor** - they average roughly half the utilization.

For Rec Center operations:

- Use predicted peak windows to **align staffing and equipment availability** with expected demand.
- Publish a simple **low / medium / high forecast** by location and hour - the classification model supports this directly.
- Consider integrating predictions into the **Rec Center app or website** as a "busyness forecast" feature.

Limitations

- Occuspace estimates are not exact headcounts; utilization above 100% suggests capacity list mismatches or sensor noise.
- The model reflects historical patterns and may not immediately capture sudden changes (new equipment, policy changes, or facility renovations).
- The model does not currently include special campus events, weather, intramural schedules, facility closures, or promotions, which may explain some one-off prediction errors.
- Predictions are most reliable for **regular recurring patterns** rather than exceptional one-time events.

Next steps

- Deploy a weekly-updated forecast dashboard by location and hour.
- Add real-time Occuspace API integration for live + predicted views.
- Re-train annually to incorporate new academic calendar years and facility changes.

This report accompanies the full reproducible analysis in the Rec Center GitHub repository (README).